IGCSE 0607 Practice Questions — E3.5

Asymptotes

Total Questions: 3 — **Total Marks:** 15 Marks

Table of Contents & Quick Navigation

- **E3.5** Asymptotes
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(2 Qs)*

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1 Let $f(x) = \dfrac{7}{x-4} + 3$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

2 A function is defined as $f(x) = \dfrac{3}{x+2} + 1$.

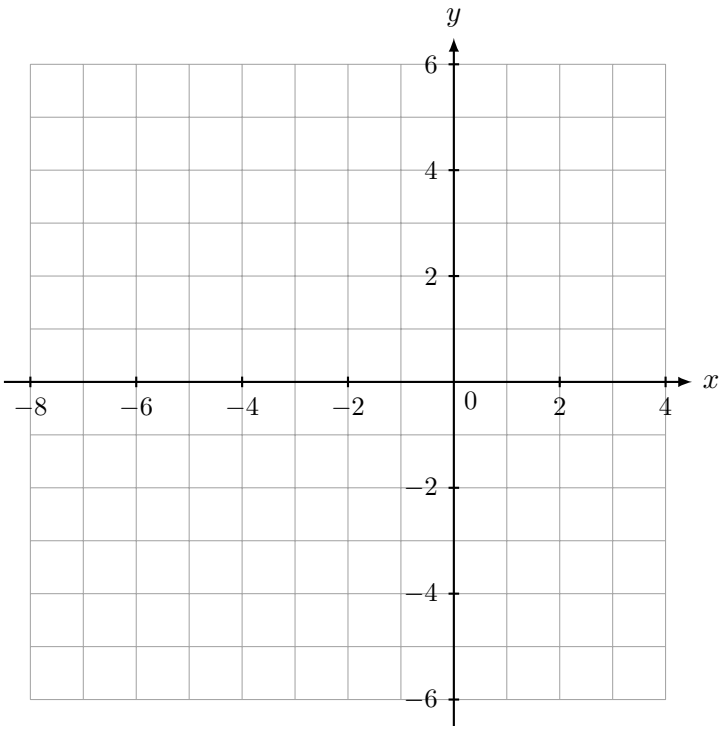
(a) Write down the equations of the two asymptotes of the graph of $y = f(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis and the y-axis.

x-axis at ..... , y-axis at ..... **[2]**

(c) On the grid, sketch the graph of $y = f(x)$.



[3]

3 A function is given by $f(x) = \dfrac{a}{x-b} + c$, where a, b and c are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = -1$.
The graph intersects the x-axis at the point $(5, 0)$.

(a) Write down the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

(c) The graph of $y = f(x)$ intersects the straight line $y = -x + 5$ at two points.
Find the x-coordinates of these two points.

$x =$ ..... , $x =$ ..... **[2]**